

Full Stack Development Internship Report

Project: Nexus – Investor & Entrepreneur Collaboration Platform

Intern: Ayesha Khan

Duration: 3 Weeks (Complete Roadmap)

Project Status: Active / Ongoing

Final Deadline: 14th April, 2026

Project Objective

The primary objective of this internship is to transform the existing Nexus frontend into a fully functional Full Stack platform. This involves integrating backend logic, secure APIs, and database management. Advanced features include Meeting Scheduling, WebRTC Video Calling, a Document Processing Chamber with e-signatures, Mock Payment Gateways, and robust Security.

Week 1 – Setup & Core Backend Foundations

Milestones & Tasks

- **Environment Setup:** Forked/cloned Nexus Repo; initialized Node.js, Express, and MongoDB environment; established frontend-backend handshake.
- **Authentication System:** Implemented JWT-based security with hashed passwords (bcrypt).
- **Role-Based Access Control (RBAC):** Created distinct workflows and dashboards for *Investors* and *Entrepreneurs*.
- **Profile Management:** Developed APIs to store and retrieve extended profile data including startup history and investment preferences.

Technical Challenges & Solutions

Challenge	Solution
CORS errors during frontend-backend connection.	Implemented <code>cors</code> middleware with specific origin whitelisting.
MongoDB Atlas connection timeouts on Railway.	Configured IP Access to <code>0.0.0.0/0</code> and optimized connection strings.
JWT Token persistence on page refresh.	Implemented <code>HTTP-Only cookies</code> /Local Storage strategy with <code>Axios</code> interceptors.

Week 2 – Collaboration & Document Handling

Milestones & Tasks

- **Meeting Scheduling:** Created a scheduling engine with conflict detection logic to prevent double-bookings.
- **Video Calling (WebRTC):** Built a signaling server using `Socket.IO` to manage peer-to-peer room connections.
- **Document Chamber:** Built a file upload system using `Multer` and `Cloudinary`; integrated `react-pdf` for in-browser previews.
- **E-Signatures:** Developed a signature capture tool (`react-signature-canvas`) that links signature images to document metadata.

Technical Challenges & Solutions

Challenge	Solution
WebRTC "Ice Candidate" failures in production.	Implemented STUN/TURN servers to bypass symmetric NAT issues.
Complexity of date-range queries for scheduling.	Utilized MongoDB \$and operators to verify overlapping UTC timestamps.
Large PDF file upload failures.	Increased Express body-parser limits and implemented multipart uploads via Cloudinary.

Week 3 – Payments, Security & Final Deployment

Milestones & Tasks

- **Payment Integration:** Developed mock transaction endpoints using Stripe/PayPal Sandbox for deposits and transfers.
- **Advanced Security:** Added input sanitization (preventing XSS/SQLi) and a 2FA mockup using Nodemailer for OTP verification.
- **API Documentation:** Generated full documentation via Swagger/Postman to facilitate testing.
- **Final Deployment:** Frontend deployed on Vercel; Backend deployed on Render/Railway.

Technical Challenges & Solutions

Challenge	Solution
Webhook verification for Stripe events.	Used <code>stripe-cli</code> for local testing and secure signature verification in the backend.
Environment Variable mismatch in production.	Standardized naming conventions (e.g., <code>VITE_</code>) across Vercel and Render settings.
Slow initial load on Render free tier.	Implemented a "keep-alive" ping script and optimized API response sizes.

Final Deliverables Summary

By the conclusion of this 3-week sprint, the following will be delivered:

- ✓ **Functional Web App:** Complete integration of Auth, Calendar, Video, Documents, and Payments.
- ✓ **Repositories:** Clean, documented GitHub repository (Frontend + Backend).
- ✓ **Deployments:** Production-ready live links for both client and server.
- ✓ **Documentation:** Detailed API docs and a final demo presentation.

Conclusion

Despite the technical hurdles—ranging from deployment networking to complex real-time signaling—the project is on track for the April 14th deadline. The current version of Nexus successfully bridges the gap between static UI and a dynamic, collaborative ecosystem.